

Formalization notes for Erdős Problem 524 in Lean 4

Companion memo to the branch `add-erdos-524` of
`leanprover-community/formal-conjectures`

Kieran McShane, with Claude Opus 4.7

April 23, 2026

Abstract

This memo accompanies the Lean 4 formalization of Chojecki’s (2026) resolution of Erdős Problem 524 — the almost-sure order of magnitude of the supremum on $[-1, 1]$ of a random algebraic polynomial with Rademacher coefficients. It records (a) the mathematical architecture of the proof as formalized, (b) precise statements of the sorry-free closed pieces, (c) the atomic set of axioms used to stand in for Mathlib-level prerequisites (Brownian motion, Itô integral, Gao–Li–Wellner, 2D KMT, endpoint reparametrization), and (d) mathematical subtleties and open concerns uncovered in the course of formalization. The memo is updated as the branch progresses. Cross-references **typeset in this font** point to actual Lean names in `FormalConjectures/ErdosProblems/`.

Contents

1	The problem and its resolution	2
1.1	Erdős’s original question and the Salem–Zygmund clarification	2
1.2	Chojecki’s theorem (January 2026)	2
2	Formalization architecture	2
2.1	Six helper modules	3
3	The LIL backbone (sorries #2, #3, #4)	3
3.1	Sharp Rademacher lower tail at the LIL scale (sorry #2)	3
3.2	Block Borel–Cantelli (sorry #3)	4
3.3	Kolmogorov’s LIL lower bound (sorry #4)	4
4	The sparse lower envelope (sorry #5)	5
4.1	Atomic sub-axioms	5
4.2	Assembly of Theorem 2	6
5	Subtleties and open concerns	7
6	Change log	8

1 The problem and its resolution

1.1 Erdős's original question and the Salem–Zygmund clarification

Let $t \in (0, 1)$ have binary expansion $t = \sum_{k \geq 1} \varepsilon_k(t) 2^{-k}$ with $\varepsilon_k(t) \in \{0, 1\}$, and set $a_k(t) := (-1)^{\varepsilon_k(t)} \in \{-1, +1\}$. Under Lebesgue measure on $(0, 1)$ the sequence $(a_k)_{k \geq 1}$ is i.i.d. Rademacher. The associated random algebraic polynomial is

$$P_n(t)(x) := \sum_{k=1}^n a_k(t) x^k,$$

and Erdős asked for the correct order of magnitude of

$$M_n(t) := \max_{x \in [-1, 1]} |P_n(t)(x)|.$$

(Per Salem–Zygmund 1954 the supremum is taken over $x \in [-1, 1]$ with Rademacher ± 1 coefficients; Erdős's original statement used binary digits $\{0, 1\}$ and $x \in [0, 1]$, which is equivalent modulo reparametrization.)

1.2 Chojecki's theorem (January 2026)

Chojecki's paper establishes two complementary results.

Theorem 1 (Upper envelope, Chojecki 2026). *Almost surely,*

$$\limsup_{n \rightarrow \infty} \frac{M_n(\omega)}{\sqrt{2n \log \log n}} = 1.$$

Theorem 2 (Sparse lower envelope, Chojecki 2026 Theorem 18). *Let $n_m := \lfloor e^{m^3} \rfloor$. There exist explicit constants $\alpha_{\pm} = (1/(6c_{\mp}))^{1/3}$, where $c_- \leq c_+$ are the Gao–Li–Wellner small-deviation constants for the centered Gaussian process $Y(u) = \int_0^1 e^{-us} dB(s)$, $u \geq 0$, such that almost surely*

$$\alpha_- \leq \limsup_{m \rightarrow \infty} \frac{\log(\sqrt{n_m}/M_{n_m}(\omega))}{(\log \log n_m)^{1/3}} \leq \alpha_+.$$

Equivalently, $M_{n_m}(\omega) = \sqrt{n_m} \cdot \exp(-\Theta((\log \log n_m)^{1/3}))$ infinitely often.

The *exact* asymptotic constant α_* in $\lim_m \log(\sqrt{n_m}/M_{n_m})/(\log \log n_m)^{1/3} = \alpha_*$ is open (Remark 19, Chojecki); it would require the as-yet-unestablished sharp one-point small-ball asymptotic for Y .

2 Formalization architecture

The Lean file 524.lean declares six sorries:

#1 wrapper, #2 LIL lower tail, #3 block BC, #4 Kolmogorov LIL, #5 sparse lower env., #6 exact constant.

All but #6 are closed (some conditionally on axioms documenting Mathlib-level gaps).

2.1 Six helper modules

Under `FormalConjectures/ErdosProblems/Helpers/` we build:

<code>StirlingTwoSided</code>	Effective two-sided Stirling bound with absolute constants (no Robbins error for the upper direction, which is sufficient downstream).
<code>IndepSetBridge</code>	Bridge lemma: $\text{iIndepFun} \Rightarrow \text{iIndepSet}$ for preimage events.
<code>BlockIndep</code>	Independence of sums over pairwise disjoint finset blocks from an i.i.d. sequence.
<code>LilNormAsymptotics</code>	Asymptotic ratios for the LIL norm $\sqrt{2n \log \log n}$ along geometric subsequences $\lfloor c^k \rfloor$.
<code>CentralBinomLower</code>	Sharp pointwise lower bound on the central binomial PMF $\binom{n}{k}/2^n$ with exponent $\alpha = 2 + \varepsilon$ (regime $(k - n/2)^2 \leq n \log n$).
<code>CentralBinomWindowSum</code>	Existential lower bound on the sum of $\binom{n}{k}/2^n$ over a window of width $\lfloor \sqrt{n} \rfloor$ centered above the LIL threshold.
<code>CubicSubseqAsymptotics</code>	Asymptotics of the cubic subsequence $n_m := \lfloor e^{m^3} \rfloor$ used in Theorem 2.

The core probabilistic content lives in `524.lean` itself.

3 The LIL backbone (sorries #2, #3, #4)

The upper envelope (Theorem 1) reduces to a functional LIL for the simple symmetric random walk

$$S_n(\omega) := \sum_{k=1}^n a_k(\omega), \quad \phi(n) := \sqrt{2n \log \log n}.$$

Classical Kolmogorov's LIL states $\limsup_{n \rightarrow \infty} S_n/\phi(n) = 1$ a.s. Chojecki's paper uses this as a building block; Mathlib lacks the $\limsup \geq 1$ direction, so we formalize it from first principles.

3.1 Sharp Rademacher lower tail at the LIL scale (sorry #2)

Theorem 3 (`lil_tail_lower_at_scale`, closed). *For every $\delta \in (0, 1)$ there exist $N(\delta) \in \mathbb{N}$ and $C(\delta) > 0$ such that for all $n \geq N(\delta)$ and every i.i.d. Rademacher sequence a on any probability space,*

$$\mathbb{P}(S_n \geq (1 - \delta)\phi(n)) \geq C(\delta) \cdot (\log n)^{-((1-\delta)^2 + \delta)}.$$

The $+\delta$ slack in the exponent makes the tails summable along geometric subsequences $n_k = \lfloor c^k \rfloor$ only for c large enough, which is exactly what the second Borel–Cantelli needs.

Proof architecture. The PMF of S_n is

$$\mathbb{P}(S_n = 2k - n) = \binom{n}{k} 2^{-n}, \quad 0 \leq k \leq n,$$

formalized as `walk_pmf_binomial` via the cylinder-event expansion of $\{a_j = \sigma_j\}_j$ for $\sigma \in \{\pm 1\}^n$ with $\sum \sigma_j = 2k - n$. Writing $d := k - n/2$, a second-order Taylor expansion of $y \mapsto (1 + y) \log(1 + y) + (1 - y) \log(1 - y)$ yields the *sharp* Kullback–Leibler bound

$$k \log \frac{2k}{n} + (n - k) \log \frac{2(n - k)}{n} \leq \frac{(k - (n - k))^2}{2n} \cdot \left(1 + O\left(\frac{d^2}{n^2}\right)\right) = \frac{2d^2}{n} \cdot (1 + o(1)). \quad (1)$$

Combined with two-sided Stirling, this produces the pointwise PMF estimate

$$\binom{n}{k} 2^{-n} \geq \frac{c}{\sqrt{n}} \exp\left(-(2+\varepsilon)\frac{d^2}{n}\right) \quad \text{whenever } d^2 \leq n \log n, \ n \geq N(\varepsilon). \quad (2)$$

Summing over a window $k \in [k_*, k_* + W]$ with $k_* = \lceil (n+t)/2 \rceil + 2$, $W = \lfloor \sqrt{n} \rfloor$, $t = (1-\delta)\phi(n)$, gives, for every $\omega \in \{S_n \geq t\}$ included via $\{S_n = 2k - n\}$ with $2k - n \geq t$:

$$\mathbb{P}(S_n \geq t) \geq W \cdot \frac{c}{\sqrt{n}} \cdot \exp\left(-(2+\varepsilon) \max_{k \in [k_*, k_*+W]} \frac{d_k^2}{n}\right).$$

The worst-case exponent expands as $(t/2 + \sqrt{n} + 2)^2 = t^2/4 + t(\sqrt{n} + 2) + (\sqrt{n} + 2)^2$, and dividing by n and invoking the algebraic core

$$(2 + \varepsilon/4) \cdot (t/2 + \sqrt{n} + 2)^2/n \leq ((1-\delta)^2 + \delta) \log \log n$$

(encapsulated as `Erdos524.Helpers.exponent_window_bound_uv`) produces the claim.

Remark 4 (Why $\alpha = 4$ is provably insufficient). If (1) is replaced by the loose first-order bound $\log x \leq x - 1$, the PMF exponent degrades from $2 + \varepsilon$ to 4. The resulting lower bound on $\mathbb{P}(S_n \geq t)$ then produces exponent $2(1-\delta)^2 \log \log n$, which *exceeds* the target $(1-\delta)^2 + \delta$ whenever $2(1-\delta)^2 > (1-\delta)^2 + \delta$, i.e. $(1-\delta)^2 > \delta$, i.e. $\delta < (3-\sqrt{5})/2 \approx 0.382$. For these δ the $+\delta$ slack cannot absorb the factor-of-2 loss. The second-order refinement of the KL bound is therefore *necessary*, not cosmetic. In Lean, the key step that unblocks it is keeping $u = \sqrt{2s}$ as a hypothesis $u^2 = 2s$ so that the core inequality (`margin_absorbs_sqrt_core`) is polynomial in (u, s, δ) — three earlier agent attempts blew the `maxHeartbeats` budget working directly with `Real.sqrt`.

3.2 Block Borel–Cantelli (sorry #3)

Theorem 5 (`lil_sparse_lower_bc`, closed). *For $\delta \in (0, 1)$ and $c > 1$, almost surely*

$$S_{\lfloor c^{k+1} \rfloor} - S_{\lfloor c^k \rfloor} \geq (1-\delta)\phi(\lfloor c^{k+1} \rfloor - \lfloor c^k \rfloor) \quad \text{infinitely often.}$$

Proof architecture. By shift-invariance of the Rademacher sequence, the increment over the block $(\lfloor c^k \rfloor, \lfloor c^{k+1} \rfloor]$ has the same law as S_{m_k} with $m_k = \lfloor c^{k+1} \rfloor - \lfloor c^k \rfloor$. Theorem 3 (applied uniformly to *every* Rademacher sequence by the strengthened signature $\exists N C, \forall a \dots$) gives

$$\mathbb{P}(E_k) \geq C \cdot (\log m_k)^{-((1-\delta)^2 + \delta)} \sim C \cdot (k \log c)^{-((1-\delta)^2 + \delta)},$$

which is not summable since $(1-\delta)^2 + \delta < 1$ for $\delta \in (0, 1)$. Disjointness of the index blocks gives `iIndepFun` of the block sums via `Erdos524.Helpers.iIndepFun_block_sums`; passing to preimage events via `iIndepSet_preimage_of_iIndepFun` yields `iIndepSet` of (E_k) , and the second Borel–Cantelli (`ProbabilityTheory.measure_limsup_eq_one`) gives the conclusion.

3.3 Kolmogorov’s LIL lower bound (sorry #4)

Theorem 6 (`kolmogorov_lil_lower_bound`, closed). *For every i.i.d. Rademacher sequence,*

$$\limsup_{n \rightarrow \infty} S_n / \phi(n) \geq 1 \quad \text{a.s.}$$

Proof. Suffices to show: for every integer $m \geq 1$, $\limsup S_n/\phi(n) \geq 1 - 1/m$ almost surely; countable intersection gives $\limsup \geq 1$ via $\lim_{m \rightarrow \infty} (1 - 1/m)$.

Fix m , set $c := 4m^2$, $\delta := 1/(8m)$. By the block BC: a.s. infinitely often,

$$S_{\lfloor c^{k+1} \rfloor} - S_{\lfloor c^k \rfloor} \geq (1 - \delta)\phi(\lfloor c^{k+1} \rfloor - \lfloor c^k \rfloor).$$

Apply Kolmogorov's LIL *upper* bound (already in the file, from Chojecki's companion argument) to $-a$, yielding a.s. eventually

$$S_{\lfloor c^k \rfloor} \geq -(1 + \delta)\phi(\lfloor c^k \rfloor).$$

Therefore a.s. infinitely often,

$$\frac{S_{\lfloor c^{k+1} \rfloor}}{\phi(\lfloor c^{k+1} \rfloor)} \geq (1 - \delta) \cdot \frac{\phi(\lfloor c^{k+1} \rfloor - \lfloor c^k \rfloor)}{\phi(\lfloor c^{k+1} \rfloor)} - (1 + \delta) \cdot \frac{\phi(\lfloor c^k \rfloor)}{\phi(\lfloor c^{k+1} \rfloor)}.$$

The ratios converge (`Erdos524.Helpers.lilNormAux_block_ratio_tendsto` and `..._scale_ratio_tendsto`) to

$$\sqrt{(c-1)/c} = \sqrt{1 - 1/(4m^2)} \quad \text{and} \quad 1/\sqrt{c} = 1/(2m),$$

respectively. The limit of the right-hand side is therefore

$$(1 - \frac{1}{8m})\sqrt{1 - \frac{1}{4m^2}} - (1 + \frac{1}{8m}) \cdot \frac{1}{2m}.$$

A small algebraic calculation (closed in Lean by `nlinarith` on the identity $((1 - \delta)\sqrt{1 - 1/c} - (1 + \delta)/\sqrt{c} - (1 - 1/m)) \cdot 32m^3 = 12m^2 - 10m + 1 \geq 3 > 0$) shows this exceeds $1 - 1/m$, yielding the conclusion.

4 The sparse lower envelope (sorry #5)

Theorem 2 is the nontrivial content of Chojecki's paper. The Lean formalization reduces it to six *atomic sub-axioms* documenting distinct Mathlib-level gaps, plus an *assembly theorem* that combines them; the assembly is still a `sorry`, but with all ingredients isolated.

4.1 Atomic sub-axioms

The following are declared in `524.lean` immediately after the `GaoLiWellnerConstants` structure. Each one stands for a specific mathematical ingredient missing from Mathlib as of v4.27.0.

`wiener_process_exists`

A standard Brownian motion $B = (B_s)_{s \geq 0}$ exists on some probability space with the usual properties. Mathlib has partial Wiener-measure infrastructure via `MeasureTheory.WienerSpace`; the full construction of a continuous-path Brownian motion is ongoing work.

`ito_integral_exp_kernel`

The Itô integral $Y(u) := \int_0^1 e^{-us} dB(s)$ exists for every $u \geq 0$ and defines a centered continuous Gaussian process with explicit covariance $\mathbb{E}[Y(u)Y(v)] = (1 - e^{-(u+v)})/(u+v)$. Mathlib lacks the general Itô integral; this is a multi-year project.

`gao_li_wellner_small_ball_upper`

The upper small-ball asymptotic $\mathbb{P}(\sup_{u \geq 0} |Y(u)| \leq \varepsilon) \leq \exp(-c_- |\log \varepsilon|^3)$, where c_- is a fixed constant.

`gao_li_wellner_small_ball_lower`

The matching lower bound $\mathbb{P}(\sup_{u \geq 0} |Y(u)| \leq \varepsilon) \geq \exp(-c_+ |\log \varepsilon|^3)$. Both GLW bounds require Karhunen–Loève expansion + entropy methods on the Gaussian process Y ; again not in Mathlib.

`two_dim_KMT_coupling`

The 2D Komlós–Major–Tusnády strong invariance coupling: on the same probability space as B , there exist two processes $Z_n^\pm(u)$ coinciding in law with the reparametrized polynomials, such that

$$\sup_{u \geq 0} |Z_n^\pm(u) - Y^\pm(u)| = O(\log n / \sqrt{n}) \text{ a.s.,}$$

with $Y^\pm$ two independent copies of Y . 1D KMT itself is an open Mathlib target.

`endpoint_reparametrization`

A *deterministic* identity: the substitution $x = \pm e^{-u/n}$ sends $x \in [-1, 1] \setminus \{0\}$ to $u \in [0, \infty)$ bijectively on each sign branch, and

$$\frac{M_n(\omega)}{\sqrt{n}} = \max \left(\sup_{u \geq 0} \left| \frac{1}{\sqrt{n}} \sum_{k=1}^n a_k(\omega) e^{-uk/n} \right|, \sup_{u \geq 0} \left| \frac{1}{\sqrt{n}} \sum_{k=1}^n a_k(\omega) (-e^{-u/n})^k \right| \right).$$

This is elementary calculus; a partial Lean proof exists in `Erdos524/EndpointReparametrization`. (modulo a few cascading `Real.sqrt/biSup` rewrites still to be unstuck).

For the Borel–Cantelli assembly we also retain two *transferred* axioms stating the small-ball bounds directly for M_n after applying endpoint reparametrization + KMT + GLW:

- `polynomial_sup_small_ball_upper`: $\mathbb{P}(M_n \leq \varepsilon \sqrt{n}) \leq \exp(-c_+ |\log \varepsilon|^3)$,
- `polynomial_sup_small_ball_lower`: $\mathbb{P}(M_n \leq \varepsilon \sqrt{n}) \geq \exp(-c_- |\log \varepsilon|^3)$.

(See Section 5 Remark 8 on a naming ambiguity in these axioms.)

4.2 Assembly of Theorem 2

The assembly theorem `chojecki_sparse_lower_envelope_proof` (currently `sorry`) takes as input the transferred small-ball axioms and produces the two bounds $\alpha_- \leq \limsup R_m \leq \alpha_+$ via Borel–Cantelli on the cubic subsequence. The key cubic-asymptotic inputs, proved `sorry`-free in `CubicSubseqAsymptotics.lean`, are:

$$\log n_m / m^3 \rightarrow 1, \quad \log n_m \geq m^3 / 2 \text{ eventually,} \quad (3)$$

$$\log \log n_m / (3 \log m) \rightarrow 1, \quad \log \log n_m \geq \log m \text{ eventually.} \quad (4)$$

Upper direction: $\limsup R_m \leq \alpha_+$ **a.s.** For $\beta > \alpha_+ = (1/(6c_-))^{1/3}$, Borel–Cantelli I gives $\mathbb{P}(R_m \geq \beta \text{ i.o.}) = 0$ provided $\sum_m \mathbb{P}(R_m \geq \beta) < \infty$. Setting $\varepsilon_m := \exp(-\beta(\log \log n_m)^{1/3})$,

$$\mathbb{P}(R_m \geq \beta) = \mathbb{P}(M_{n_m} \leq \varepsilon_m \sqrt{n_m}) \leq \exp(-c_+ \beta^3 \log \log n_m) \sim m^{-3c_+ \beta^3}.$$

Summability requires $3c_+ \beta^3 > 1$, i.e. $\beta^3 > 1/(3c_+)$. The claim $\beta^3 > 1/(6c_-)$ is stronger, hence the *structural* hypothesis $c_+ \geq 2c_-$ (i.e. `two_lower_le_upper`) which we have added to the `GaoLiWellnerConstants` structure. Under this hypothesis, $\beta^3 > 1/(6c_-) \geq 1/(3c_+)$ and summability follows from (4).

Remark 7 (The factor of 6 beyond $2c_+ \leq c_-$). The true Gao–Li–Wellner bound has $c_+ \sim c_-$ asymptotically (they coincide up to a factor of 2 arising from the endpoint split, not an additional factor). Chojecki’s exact factor 6 arises as $6 = 3 \cdot 2$: the 3 from (4), and the 2 from the identity

$$\mathbb{P}(M_n \leq \varepsilon \sqrt{n}) = \mathbb{P}(M_n^+ \leq \varepsilon \sqrt{n}, M_n^- \leq \varepsilon \sqrt{n}) \approx \mathbb{P}(M_n^+ \leq \varepsilon \sqrt{n})^2$$

under the (asymptotic) independence of $M_n^\pm$. The last approximation is enforced by the KMT coupling to two independent copies of Y and is *not* a consequence of the transferred one-sided axioms as currently stated. Our structural hypothesis $c_+ \geq 2c_-$ is the minimal weakening that keeps the downstream BC1 calculation tight; a sharper formalization would either state “product-of-marginals” small-ball bounds as a separate axiom, or axiomatize the KMT step at the level of coupling to *independent* $Y^\pm$ and derive the factor of 2 internally.

Lower direction: $\liminf \limsup R_m \geq \alpha_-$ **a.s.** For $\alpha < \alpha_- = (1/(6c_+))^{1/3}$, Borel–Cantelli II requires $\sum_m \mathbb{P}(R_m \geq \alpha) = \infty$ and mutual (or pairwise) independence of the events. The transferred small-ball lower bound gives

$$\mathbb{P}(R_m \geq \alpha) \geq \exp(-c_- \alpha^3 \log \log n_m) \sim m^{-3c_- \alpha^3},$$

so divergence holds for $\alpha < (1/(3c_-))^{1/3}$ which is $\geq \alpha_-$ once the factor-of-2 independence is established (Remark 7).

The independence step is the main obstruction currently in the Lean proof; see Remark 9 below.

5 Subtleties and open concerns

Remark 8 (Naming convention of GLW constants). The `GaoLiWellnerConstants` structure declares two real fields `lower`, `upper` with the original docstring

$$\exp(-\text{upper} \cdot |\log \varepsilon|^3) \leq \mathbb{P}(\sup |Y| \leq \varepsilon) \leq \exp(-\text{lower} \cdot |\log \varepsilon|^3).$$

In this convention `upper` is the *larger* constant and drives the *lower* bound on $\mathbb{P}$; conversely `lower` is the smaller constant and drives the upper bound.

The currently declared transferred axioms `polynomial_sup_small_ball_{upper,lower}` invert this convention: the “upper” axiom uses `glw.upper` in an upper bound on $\mathbb{P}$, and vice versa. For the default instance `(lower, upper) = (1, 2)` with the freshly added `two_lower_le_upper` field, the statements remain mutually consistent, but for any semantically correct instance reflecting the GLW theorem, the axioms’ convention would need to be swapped to match the structure’s docstring. This is a clerical (not mathematical) fix and should be performed in a future commit.

Remark 9 (Block independence vs. nested supremum). The helper `iIndepFun_block_sums` gives independence of *sums* over pairwise disjoint finite-index blocks from an i.i.d. sequence. This applies directly to increments $S_{n_{k+1}} - S_{n_k}$ (used in the LIL lower bound, sorries #3–#4).

It does *not* apply directly to $M_{n_m}(\omega)$, which depends on the whole prefix $(a_1, \dots, a_{n_m})$. Independence of the events $\{M_{n_m} \leq \varepsilon_m \sqrt{n_m}\}$ in Theorem 2 is obtained via KMT: couple $M_{n_m}/\sqrt{n_m}$ to a functional of $B|_{[0,1]}$, decompose B into independent increments, truncate outside a block, and exploit that M_{n_m} is asymptotically determined by the most recent block’s Brownian increment. This is *not* a direct instance of block independence: it is a coupling-plus-truncation argument bundled into the KMT axiom, whose assembly into the actual proof is the 500-line BC content still absent from `chojecki_sparse_lower_envelope_proof`.

Remark 10 (Endpoint reparametrization as a real theorem). The sixth atomic axiom `endpoint_reparametrization` is elementary calculus and deserves elimination. A partial proof exists (`Erdos524/EndpointReparametrization.lean` not yet on `main`) containing: the substitution $u = -n \log x$, image computation $\{x \in (0, 1]\} = \{u \in [0, \infty)\}$ and the matching formula $a_k x^k \mapsto a_k e^{-uk/n}$. The final bridge into M_n (matching `randomPoly a n` ω to the helper’s `Helpers.poly a`) breaks on a `let/set` opacity issue plus a small handful of `biSup` normalization rewrites. Estimate to complete: 1–2 hours of careful Lean.

Remark 11 (Sharp LIL tail via the $u^2 = 2s$ reparametrization). The proof of (2) in Lean foundered three times on `maxHeartbeats` timeouts when tactics like `nlinarith`, `field_simp`, and `ring` were applied directly to expressions containing `Real.sqrt (2 * n * Real.log (Real.log n))`. The key tactical move (lemma `margin_absorbs_sqrt_core`) is to introduce u with a hypothesis $u^2 = 2s$ so that every inequality is *polynomial* in (u, s, δ) ; no `Real.sqrt` appears in the arithmetic. This pattern is likely useful in other LIL-scale formalizations.

Remark 12 (The `answer(sorry)` elaborator). The `FormalConjectures` framework provides `answer(_)` as a meta-syntactic placeholder; in `Prop` context with argument `sorry`, it elaborates to `True` (`FormalConjectures/Util/Answer.lean`). Thus `erdos_524 : answer(sorry)  $\leftrightarrow$  RHS` reduces to `True  $\leftrightarrow$  RHS`, i.e. proving the right-hand side directly. In our case the right-hand side asserts the existence of a single function f controlling both envelopes at the LIL scale, which is now delivered by $f(n) := \sqrt{2n \log \log n}$ and the proven $\limsup M_n/f(n) = 1$.

Remark 13 (Open problem: the exact constant (sorry #6)). The final `sorry` concerns the almost-sure *limit*

$$\lim_{m \rightarrow \infty} \frac{\log(\sqrt{n_m}/M_{n_m})}{(\log \log n_m)^{1/3}} = \alpha_*.$$

By Theorem 2 the $\limsup$ lies in $[\alpha_-, \alpha_+]$ almost surely, but the sharp one-point asymptotic $-\log \mathbb{P}(\sup |Y| \leq \varepsilon) \sim c_* |\log \varepsilon|^3$ on which $\alpha_* = (1/(6c_*))^{1/3}$ would be computed is not known; it remains an open problem in Gaussian-process small-ball theory (Chojecki 2026 Remark 19). Closing this `sorry` therefore depends on mathematical progress, not formalization effort.

6 Change log

e0af2	<code>docs</code> : update memo change log.
c6e3cba	<code>feat</code> : strengthen <code>GaoLiWellnerConstants</code> with <code>2.lower $\leq$ upper gap</code> . Default instance updated to <code>(lower, upper) = (1, 2)</code> .
68d7240	<code>feat</code> : add <code>Helpers/CubicSubseqAsymptotics.lean</code> (six theorems, <code>sorry-free</code>) and this memo.
6e7bc7e	<code>refactor</code> : decompose the opaque Chojecki axiom into six atomic Mathlib-gap sub-axioms plus two transferred axioms. Close sorries #1 and #5 (the latter conditionally on the new sub-axioms).
315e8e1	<code>feat</code> : close sorries #2, #3, #4 — full LIL lower bound chain. Add six helper modules and <code>walk_pmf_binomial</code> .

Outstanding.

- `chojecki_sparse_lower_envelope_proof` — the BC assembly on the cubic subsequence. Upper direction reduces to direct BC1 once the factor-of-2 issue (Remark 7) is handled; lower direction needs the truncation-plus-coupling argument sketched in Remark 9.
- `endpoint_reparametrization` — partially proved (a pure-calculus identity), in workspace but not yet on `main`.
- `exact_lower_constant` — open in mathematics.